Component	Qty	Voltage	Cont. Current	Stall Current	RPM	Notes	Datasheet Link	Purchase Link	Video Links
25mm JGA25-370B DC Motor with Encoder	6	6V	0.5A	2.1A	170	25mm fits in wheel (~30mm)	Specs from similar motor	Aliexpress	
DRV8871 Motor Driver	6	6.5-45V	3.6A max		--	Account for stall current	Learn Adafruit	Aliexpress	ESP32 using DRV8871
PCA9658 16-Channel Servo Driver	1	VCC: 3-5V V+: 5-12V	25 mA per pin		--	Connect to power supply to prevent burnout.	Learn Adafruit		Tutorial
ESP32 with 38 pins	1	2.7-3.6V	Average: 80 mA		--	Minimizes pin usage on Raspberry Pi	Robokreeda		
DS041MG Servos	6	7.4V	150 mA	800 mA	0.12 sec/60°	Torque: 4.0 kg	GDW Servo	Aliexpress	
Zeee LiPo Battery Pack	1	7.4V	Max Current: 260A, Capacity: 5200mAh		Discharge Rate: 50C	Need about 20A for all components peak.		Aliexpress	
Raspberry Pi 5		5V	2.5 A	Max: 5A					
LM2596 Buck Converter	2	Input: 3-40V Output:1.5-35V	2A	Max:3A		Practice for 4 motors, does create frequency noise		Aliexpress	
7805 Voltage Regulator	1					To connect to ESP32, does overheat	sparkfun		

Raberry Pi- 5: requires a 5V, 5A (25-27W) USB-C power supply for full functionality, but typical usage draws significantly less, usually peaking around 1.5A to 2.5A depending on USB peripherals.

H-Bridge: Adafruit Industries DRV8871 DC Motor Driver Breakout Board - 3.6A Max

RC Transmitter Controller

DC Motors: 25 mm DC Motor w encoder: 1.8 A per motor x 6

Articulated Robotics:
https://www.youtube.com/watch?v=_FGYVgAti9M&list=PLunhqkrRNRhYAffV8JDiFOatQXuU-NnxT&index=6

Build a low-cost Wheeled Robot Guide:

https://www.mclibre.org/descargar/docs/revistas/magpi-books/the-magpi-essentials-projects-05-en-201911.pdf

-page 156

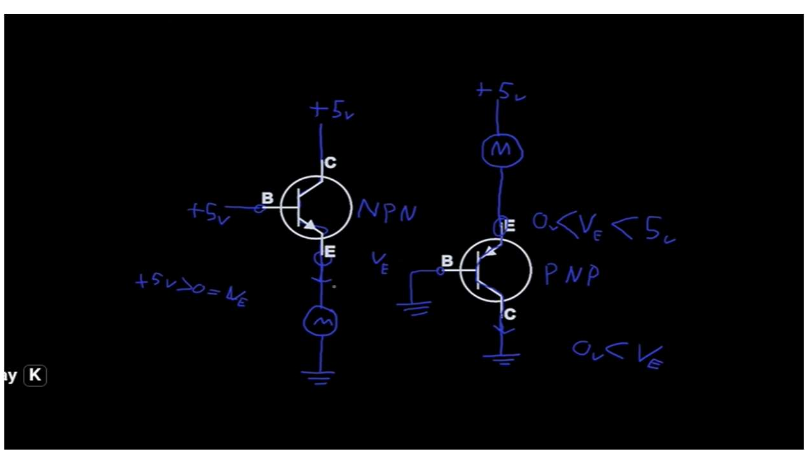
1. Transistor Basics

- Simple setup to control a DC motor.

-Uses a transistor as a switch to safely power the motor from a microcontroller or Raspberry Pi.

https://www.youtube.com/watch?v=iiv4HOnwMRg

-Wiring diagram included in the video description.



2. How H-Bridge works

-H-Bridge uses four transistors to control the direction of current.

-By changing the polarity of current flow, it allows you to spin the motor/wheels forward or backward.

-Essential for controlling DC motors in robotics (like reversing a rover's wheels).

https://www.bing.com/videos/riverview/relatedvideo?q=what+is+h-bridge+in+robotics&mid=A82EC2D7A0F206241332A82EC2D7A0F206241332&churl=https%3a%2f%2fwww.youtube.com%2fchannel%2fUC1-w1mQxEkx4K4Q3Z9_C37w&FORM=VIRE

- Wiring diagram included in the video description.

1kohm resistor

250 Kohm potentiometer

Pull Up/Pull Down Resistors

https://forum.arduino.cc/t/pull-down-resistors/417572

Driving DC Motor with Microcontroller

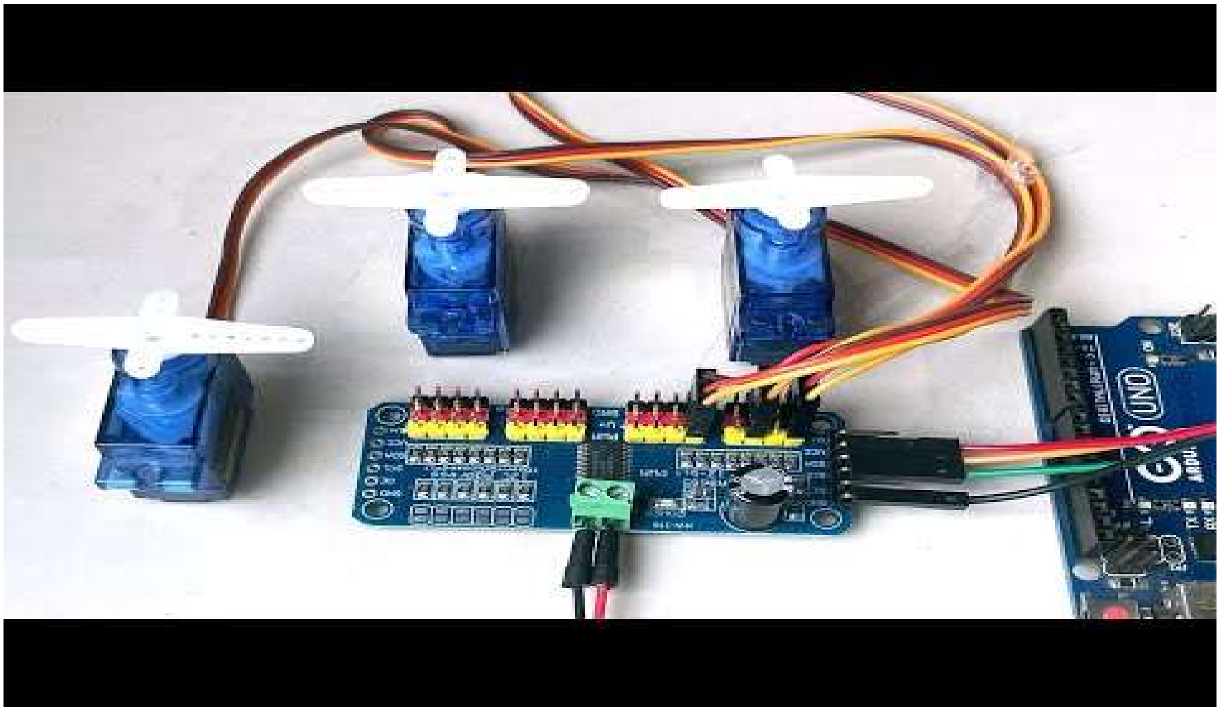
https://www.youtube.com/watch?v=ygrsIqWOh3Y

Data Sheet-JGA25-370 geared motor (245 rpm/0.65 A/1.4 Kg*cm-5.2 Kg*cm/2.2 A)

https://media.digikey.com/pdf/Data%20Sheets/Seeed%20Technology/114090046_Web.pdf

-6 motors-[normal running ~ 0.65 A x 6 = 3.9 A]

-[stall ~ 2.2 A x 6 = 13.2 A]



https://www.amazon.com/Flysky-Transmitter-Controller-Helicopter-
Quadcopter/dp/B07CWBQ2HM/ref=sr_1_1?adgrpid=185255443903&dib=eyJ2IjoiMSJ9.973jPiPGwPoLoYnBOA
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111352436446&hydadcr=4090_13514019_10970&keywords=flysky%2Bfs-
i6&mcid=ad0ae3f64a2238198994b5cef849faf2&qid=1771994333&sr=8-1&th=1

DRV8871 DC Motor Driver

https://www.microcenter.com/product/613664/drv8871-dc-motor-driver-breakout-board-36a-
max?utm_source=chatgpt.com

JGA25-370B With Hall encoder (Looks good and should fit) 12V/170 RPM

https://www.aliexpress.us/item/3256806027674487.html?src=google&pdp_npi=4%40dis%21USD%215.09%2
15.09%21%21%21%21%21%40%2112000050795094780%21ppc%21%21%21&src=google&albch=shopping&a
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https://www.aliexpress.us/item/3256804627123083.html?gatewayAdapt=glo2usa4itemAdapt

https://www.google.com/search?sca_esv=04572e48b25069e1&rlz=1CDGOYI_enUS1141US1141&hl=en-US&sxsrf=ANbL-n7296f0reQLRhioJugSKwMXmM1BpQ:1771567925909&q=brushed+dc+motors+with+encoders&source=lnms&fbs=ADc_l-YQanUcJSoe62luYRIM6gsUt2zjmW_MvZe6pHkYHWOdy8woxLkmF_YUe3IvdgohA_-59XP-JOh2ERPM_OobCFIzwLzx9_-eMFyvEokh83QF0KO66AwtQ7wjITxbIwflm_QdEcC9GlG_e-KR4lFEFeyImdCc8qxKOfTxGBsA8TI0Mo1B0kirGExGPwPtXJPd-LlUi7J2ndGw1iH5Lnm8Et8J8rrjyDgEDWEoq-yxvJDlf30tydM&sa=X&ved=2ahUKEwilkq3ntOeSAxUSJ0QIHUhcGF8Q0pQJegQICRAB&biw=440&bih=766&dpr=3

Stair Climbing Robot Jaypee Uni

http://www.ir.juit.ac.in:8080/jspui/bitstream/123456789/6517/1/Stair%20Climbing%20Robot.pdf

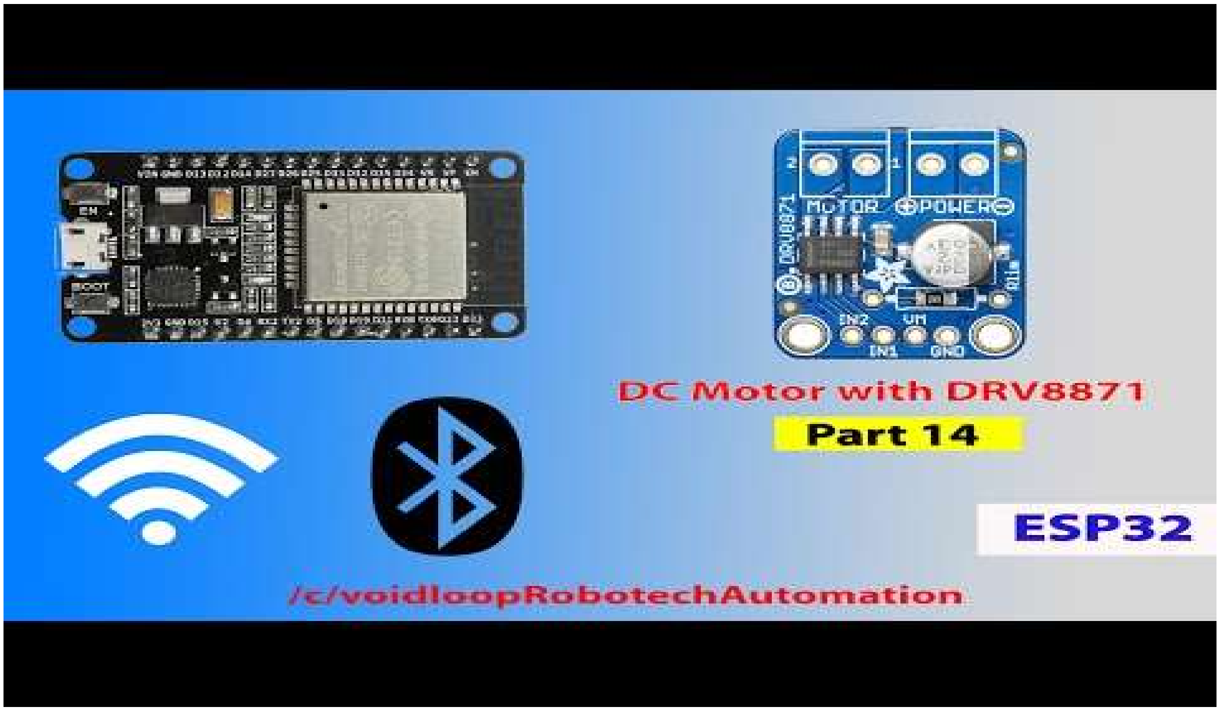
Wheel dimensions: diameter outer well-> 8 cm

-diameter DC motor well-> ~3 cm-> ~30 mm

DRV8871 Motor Driver (single motor)

-Datasheet: https://cdn-learn.adafruit.com/downloads/pdf/adafruit-drv8871-brushed-dc-motor-driver-breakout.pdf

-video https://www.youtube.com/watch?v=DIt_Y3xbBrE



PCA9658 16-channel servo driver

-Datasheet: https://cdn-learn.adafruit.com/downloads/pdf/16-channel-pwm-servo-driver.pdf

-tutorial video: https://www.youtube.com/watch?v=_DgLt2Inr1E

ESP32 with 38 pin